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# Nephrolithiasis and Nephrocalcinosis

*Wei Chen, David A. Bushinsky*

## NEPHROLITHIASIS

### Epidemiology

Kidney stones are common in high-income countries, with an annual incidence of more than 1 in 1000 persons and a lifetime risk for forming stones of approximately 9% in women and 13% in men.[1-4] In the United States the prevalence of nephrolithiasis increased from 3.2% in the late 1970s to 5.2% in the 1990s, to 8.8% in the first decade of the 2000s, and further to approximately 10% in the 2010s, in parallel with the rising incidence of obesity, insulin resistance, and type 2 diabetes mellitus.[5-7] Factors that determine kidney stone prevalence include age, sex, race, and geographic distribution. Incidence peaks in the third and fourth decades, and prevalence increases with age until approximately 70 years in men and 60 years in women. Men are more prone to stone formation than women.[5] In the United States, Whites are more likely to develop kidney stones than African Americans, Hispanics, or Asian Americans, but the prevalence is rising in non-Whites as well.[5] The tendency in the United States for the development of stones also depends on geographic location, with an increasing prevalence from north to south and, to a lesser degree, from west to east. The increase in nephrolithiasis rates from north to south may be due to increased environmental temperatures, and greater sunlight exposure leading to an increase in insensible losses through sweating and more concentrated urine.[1,8] The higher urine calcium excretion in a smaller urine volume will increase the risk for supersaturation for calcium-containing crystals, thereby promoting stone formation.

Stone type varies with worldwide geography and genetic predisposition. In the Mediterranean and Middle East, 75% of stones are composed of uric acid. In the United States, the majority of stones are calcium oxalate and/or calcium phosphate (>70%), with less than 10% being pure uric acid stones. Magnesium ammonium phosphate (struvite) stones account for 10% to 25% of stones formed (with a higher incidence in the United Kingdom), and cystine stones constitute 2% of all stones formed (Fig. 60.1).[9]

Kidney stones are associated with systemic conditions including chronic kidney disease (CKD), cardiovascular disease, and bone disease. In a Canadian cohort,[10] having one or more episodes of kidney stones was associated with increased risk for end-stage kidney disease (ESKD; adjusted hazard ratio [HR], 2.16; 95% confidence interval [CI], 1.79–2.62) and development of CKD (HR, 1.74; 95% CI, 1.61–1.88). The risk for CKD development was greater in women than in men and in people younger than 50 years. However, the absolute increase in the rate of adverse kidney outcomes associated with kidney stones was small; the unadjusted rate of ESKD was 2.48 and 0.52 per million person-days in people who developed kidney stones and in people without stones, respectively. Development of kidney stones was associated with increased risk for coronary heart disease in women but not in men, and with greater aortic calcification.[11,12] Those with calcium-based stones were shown to have lower bone mineral density and higher risk of fracture than non–stone formers.[12-14]

### Pathogenesis

Stones occur in urine that is supersaturated with respect to the ionic constituents of the specific type of stone. Supersaturation depends on the product of the free ion activities of stone components rather than on their molar concentrations. The free ion activities of stone components can be affected by the concentration of the crystal components, presence of inhibitors, and urine pH. An increasing concentration of the components of the crystal increases their free ion activity. When calcium and oxalate are dissolved in pure water, the solution becomes saturated when the addition of any more calcium or oxalate does not result in further dissolution. However, urine, unlike pure water, contains numerous calcification inhibitors that can form soluble complexes with the ionic components of the stone. The interactions with these inhibitors (e.g., citrate) may result in a decrease in free ion activity that allows the stone constituents to increase in total concentration to levels that would normally cause stone formation in pure water. Urinary pH also influences free ion activity. The level of chemical free ion activity in which stones will neither grow nor dissolve is referred to as the *equilibrium solubility product,* or the upper limit of metastability. Above this level, the urine will be supersaturated, and any stone present will grow in size.

When the solution becomes supersaturated with respect to a solid phase, ions can join together to form the more stable, solid phase, a process termed *nucleation.* Homogeneous nucleation refers to the joining of similar ions into crystals. The more common and thermodynamically favored heterogeneous nucleation results when crystals grow adjacent to crystals or other substances in the urine, such as sloughed epithelial cells. Calcium oxalate crystals, for example, can nucleate with uric acid crystals. These small crystals may then aggregate to form larger stones, which would pass into the urine, causing crystalluria, if they did not anchor to the urothelium.

Calcium oxalate crystals anchor on areas of calcium phosphate deposits termed *Randall plaques,* which are located in the renal papillae. Randall plaques appear to originate around the thin loop of Henle. In the outer medulla, the vascular bundles are surrounded by the thick ascending limb, which absorbs calcium independently of water. As the delivery of calcium to the thick ascending limb increases, the calcium concentration in the vascular bundles increases. This promotes supersaturation of crystals in the thin limb, which fosters plaque formation.

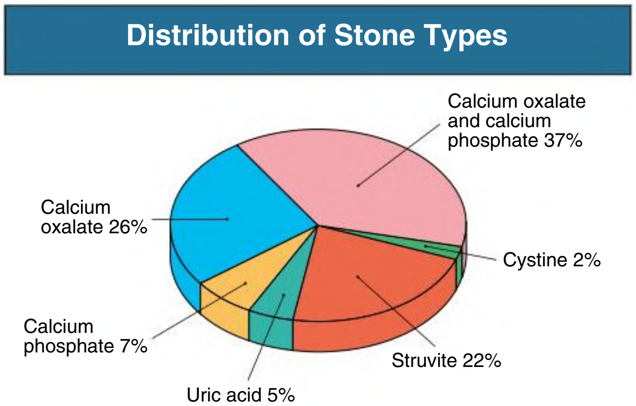


**Fig. 60.1** Proportion of stone types in a typical U.S. population.

**Fig. 60.2 Ureteral Calculus.** A 1-cm-wide calcium oxalate stone that provoked ureteral colic and required surgical removal.

**TABLE 60.1 Clinical Presentations of Nephrolithiasis**

| Presentation | Characteristics |
|---|---|
| Pain | Ureteral colic, loin pain, dysuria |
| Hematuria | – |
| Urinary tract infection | Recurrent, chronic infection, pyelonephritis |
| Asymptomatic urine abnormality | Microhematuria, proteinuria, sterile pyuria |
| Interruption of urinary stream | – |
| Calculus anuria | – |

**BOX 60.1 Causes of Hematuria**

- Nephrolithiasis
- Infection: cystitis, prostatitis, urethritis, acute pyelonephritis, tuberculosis, schistosomiasis
- Malignancy: renal cell carcinoma, transitional cell carcinoma, prostate cancer, Wilms tumor
- Trauma
- Glomerular disease
- Interstitial nephritis
- Polycystic kidney disease
- Papillary necrosis
- Medullary sponge kidney
- Miscellaneous: loin pain–hematuria syndrome, arteriovenous malformation, chemical cystitis, caruncle, factitious

This theoretical mechanism is referred to as *vas washdown* and has yet to be proven experimentally.[15] Calcium oxalate crystals attach to Randall plaques, allowing significant stone growth.[16] Clinically apparent stone disease occurs when the calcium oxalate crystals break off from the Randall plaques and cause injury or obstruction.

Several genetic polymorphisms have been implicated in the pathogenesis of calcium stones in genome-wide association studies. They include genes coding for proteins regulating tubular calcium and phosphate reabsorption (e.g., calcium-sensing receptor, vitamin D receptor, and claudins), genes coding for proteins preventing calcium salt precipitation (e.g., matrix Gla protein), and genes coding for aquaporin in the proximal tubule.[3,17] Besides genetic polymorphisms, microbiome and exposure to antibiotics have also been implicated in the pathogenesis of nephrolithiasis. Exposure to sulfas, fluoroquinolones, cephalosporins, nitrofurantoin/methenamine, and broad-spectrum penicillin was associated with increased odds of nephrolithiasis.[18,19] Antibiotics may change the composition of the intestinal microbiome and alter the metabolism of macronutrients and ionic absorption, thus affecting the risk of nephrolithiasis.

## Clinical Manifestations

The two most characteristic symptoms of nephrolithiasis are pain and hematuria. Other presentations include urinary tract infection (UTI) and acute kidney injury caused by obstructive uropathy if stones cause bilateral kidney outflow tract obstruction or unilateral obstruction in a single functioning kidney (Table 60.1).

### Pain

The classic manifestation of pain in patients with nephrolithiasis is ureteral colic. Pain is of abrupt onset and intensifies over time into an excruciating, severe flank pain that resolves only with stone passage or removal. The pain may migrate anteriorly along the abdomen and inferiorly to the groin, testicles, or labia majora as the stone moves toward the ureterovesical junction. Gross hematuria, urinary urgency, frequency, nausea, and vomiting may occur. Stones smaller than 5 mm usually pass spontaneously with hydration, whereas progressively larger stones are increasingly likely to require urologic intervention (Fig. 60.2).[16] Ureteral colic may occur with the passage of clots from hematuria of any cause ("clot colic") or with papillary necrosis. Nephrolithiasis also may provoke less-specific loin pain that poorly localizes to the kidney and therefore has a wide differential diagnosis, particularly if not associated with other urinary symptoms. The finding of a stone on radiologic examination does not preclude a coincidental cause of pain from another source.

### Hematuria

Stone disease is a common cause of hematuria. Macrohematuria occurs more commonly with large calculi and during UTI and colic. Although typically associated with loin pain or ureteral colic, the hematuria of nephrolithiasis also may be painless. The clinical differential diagnosis of hematuria is wide (Box 60.1). Painless microhematuria in children may occur with hypercalciuria in the absence of demonstrable stones.

### Loin Pain–Hematuria Syndrome

Loin pain–hematuria syndrome is a poorly understood condition that must be considered in the differential diagnosis of nephrolithiasis. It is diagnosed by exclusion when patients (most typically young and middle-aged women) present with loin pain and persistent microhematuria or intermittent macrohematuria.[20] Careful evaluation is required to exclude small stones, tumor, UTI, and glomerular disease.

Angiographic abnormalities implying intrarenal vasospasm or occlusion have been reported, as have kidney biopsy abnormalities typified by deposition of complement C3 in arteriolar walls. However, these findings are not consistent, nor do they provide a coherent framework to explain the pathogenesis of this condition.

In one study, 43 consecutive patients with clinical manifestations of loin pain–hematuria syndrome were evaluated by kidney biopsy after other causes of their symptoms had been excluded with at least two imaging studies.[20] Thirty-four patients were considered to have idiopathic loin pain–hematuria syndrome after nine with histologic evidence of immunoglobulin A nephropathy were excluded. Of these, 66% had glomerular basement membranes that were either unusually thick or thin on electron microscopy and 47% had a history of kidney stones, though none had obstructing stones at the time of imaging assessment. Evidence of glomerular hematuria was more common in biopsy samples of patients with loin pain–hematuria syndrome compared with those of healthy living kidney donors who also underwent kidney biopsy. The investigators postulated that the structurally abnormal glomerular basement membranes in the majority of these patients may lead to rupture of the glomerular capillary walls, with consequent hemorrhage into the renal tubules and tubular obstruction by red blood cells, triggering kidney edema, stretching of the renal capsule, and severe flank pain.

Loin pain–hematuria syndrome is a chronic condition requiring reassurance, careful management of analgesia, and ongoing psychological support. The condition usually remits after several years. Denervation of the kidney by autotransplantation is rarely successful. The extreme measure of nephrectomy has been used, but pain often recurs promptly in the contralateral kidney. Bilateral nephrectomy and kidney replacement therapy have been reported as an approach of very last resort. Referral to a pain clinic can assist in providing psychiatric counseling, analgesia, and exclusion of other disorders. In one retrospective study, patients who eventually came to accept a nonsurgical approach along with pain-coping strategies that did not involve narcotic analgesics had the most successful outcomes.[21]

### Asymptomatic Stone Disease

Even large staghorn calculi may be asymptomatic and discovered only during the investigation of unrelated abdominal or musculoskeletal symptoms. Obstructive uropathy caused by calculi also may be painless; therefore, nephrolithiasis always should be considered in the differential diagnosis of unexplained kidney failure. In an outbreak of melamine-associated nephrolithiasis in Chinese infants, the majority of patients brought to a screening clinic had no symptoms or signs of stones, and the diagnosis of nephrolithiasis was made only by ultrasound in at-risk infants and toddlers.[22]

## Clinical Evaluation of Stone Formers

All patients with recurrent nephrolithiasis merit metabolic evaluation to determine the cause of their kidney stones.[23] Patients with a single stone should at least undergo a basic evaluation. Those with metabolically active stones (stones growing in size or number within 1 year), all children, non–calcium stone formers, and patients in demographic groups not typically prone to stone formation warrant a complete evaluation, which includes a 24-hour urine collection made with the patient ingesting his or her typical diet.

### Basic Evaluation

The evaluation of stone formers includes a careful history and physical examination and requires specific data gathering on stone formation, diet, and specific laboratory studies, as shown in Box 60.2.

**BOX 60.2 Basic Evaluation of Nephrolithiasis**

- Stone history
  - Number of stones formed
  - Frequency of stone formation
  - Age at first onset
  - Size of stones passed or still present
  - Kidney involved (left, right, or both)
  - Stone type, if known
  - Need for urologic intervention such as extracorporeal shock wave lithotripsy or percutaneous nephrolithotomy
  - Response to surgical procedure
  - Association of stones with urinary tract infections
- Medical history
- Medications
- Family history
- Occupation and lifestyle
- Fluid intake and diet
- Physical examination
  - Evidence of systemic causes of stones (e.g., tophi)
- Laboratory data
  - Urinalysis
  - Urine culture
  - Stone analysis
- Blood chemistry
  - Sodium, potassium, chloride, bicarbonate
  - Creatinine, calcium, phosphorus, uric acid
  - Intact parathyroid hormone level if calcium elevated
- Radiologic evaluation
  - Abdominal radiograph (no contrast)
  - Noncontrast computed tomography
  - Ultrasound (generally recommended)

***History.*** The history may uncover a systemic cause for nephrolithiasis. Any disease that can lead to hypercalcemia (including malignancy, hyperparathyroidism, and sarcoidosis) can result in hypercalciuria and increase the risk for calcium stone formation. A number of malabsorptive gastrointestinal disorders (including Crohn disease and celiac disease) can result in calcium oxalate stone formation as a result of volume depletion and hyperoxaluria. Uric acid stones often occur in patients with a history of gout and insulin resistance.[9]

The stone history (see Box 60.2) includes the number and frequency of stones formed, patient age at incidence of first stone, size of stones, stone type (if known), and whether the patient required surgical removal of the calculi. This information indicates the severity of the stone disease and provides clues to the cause of the stone formation. For example, large staghorn calculi that do not pass spontaneously and recur despite frequent surgical intervention are more consistent with struvite than calcium oxalate stones. Stones that develop at a young age may be caused by cystinuria or primary hyperoxaluria. Stone response to intervention is also significant; cystine stones, for example, do not fragment well with lithotripsy. If stones recur frequently in a single kidney, a congenital abnormality in that kidney, such as megacalyx or medullary sponge kidney, should be explored.

Family history is important because a number of stone types have a genetic basis. Idiopathic hypercalciuria appears to be a polygenic disorder. Mutations in the claudins, which regulate calcium reabsorption in the thick ascending limb of the loop of Henle, cause familial hypercalciuria and nephrocalcinosis. A genome-wide association study in

## BOX 60.3 Medications Associated With Nephrolithiasis and Nephrocalcinosis

### Calcium Stone Formation
- Loop diuretics
- Vitamin D
- Corticosteroids
- Calcium supplements
- Antacids (calcium and noncalcium antacids)
- Theophylline
- Acetazolamide[a]
- Amphotericin[a]
- Topiramate

### Uric Acid Stone Formation
- Salicylates
- Probenecid
- Melamine (in contaminated infant formula and milk products)

### Medications That May Precipitate Into Stones
- Triamterene
- Acyclovir (if infused rapidly intravenously)
- Indinavir
- Nelfinavir

[a]Associated with nephrocalcinosis.

## BOX 60.4 Foods High in Oxalate and Purine

### High-Oxalate Foods (Avoid in Setting of Hyperoxaluria)
- Green beans
- Beets
- Celery
- Green onions
- Leeks
- Leafy greens: collard greens, dandelion greens, Swiss chard, spinach, escarole, mustard greens, sorrel, kale, rhubarb
- Cocoa
- Chocolate
- Black tea
- Berries: blackberries, blueberries, strawberries, raspberries, currants, gooseberries
- Orange peel
- Lemon peel
- Dried figs
- Summer squash
- Nuts, peanut butter
- Tofu (bean curd)

### High-Purine Foods (Avoid in Setting of Hyperuricosuria)
- Organ meats: sweetbreads, liver, kidney, brains, heart
- Shellfish
- Meat: beef, pork, lamb, poultry
- Fish: anchovies, sardines (canned), herring, mackerel, cod, halibut, tuna, carp
- Meat extracts: bouillon, broth, consommé, stock
- Gravies
- Certain vegetables: asparagus, cauliflower, peas, spinach, mushrooms, lima and kidney beans, lentils

patients with kidney stones identified sequence variants in the gene encoding claudin 14 that were associated with hypercalciuria.[24]

Cystinuria is usually autosomal recessive, and hyperuricosuria has been associated with rare inherited metabolic disorders. Nephrolithiasis and nephrocalcinosis can result from a variety of monogenic disorders, such as Dent disease (X-linked recessive nephrolithiasis), McCune-Albright syndrome, osteogenesis imperfecta type 1, and congenital lactate deficiency (see also Chapter 50). The various genetic disorders can lead to hypercalciuria by increasing bone resorption, affecting intestinal absorption, by decreasing renal tubular reabsorption transport, or by unknown mechanisms.[1,16,24]

Certain medications can potentiate calcium stone formation (e.g., loop diuretics are calciuric) or may predispose to uric acid lithiasis (salicylates, probenecid) (Box 60.3). Drugs can also precipitate into stones themselves, such as rapidly infused intravenous acyclovir, high-dose sulfadiazine, triamterene, and the antiretroviral agents indinavir and nelfinavir.[25] In addition, some medications (e.g., acetazolamide, topiramate) promote nephrolithiasis by inhibiting carbonic anhydrase activity. With these drugs, the metabolic acidosis that ensues, along with lower urine citrate, higher urine pH, and increased urinary calcium excretion, predisposes to calcium phosphate stone formation.

The social history should include details regarding occupation and lifestyle. Surgeons and real-estate agents, for example, may minimize fluid intake to avoid bathroom breaks during the workday. Those who engage in vigorous physical activities, such as running, may not rehydrate adequately to keep up with insensible losses, producing excessively concentrated urine and precipitation of stone crystals in those prone to nephrolithiasis.

A dietary history and review of fluid intake are essential in determining potential causes or contributors to stone formation. The patient should be asked about commonly consumed foods, with attention paid to sodium-containing foods, as well as quantities of calcium, animal protein, purine, and oxalate (Box 60.4). Dietary calcium intake should be reviewed, because many patients with nephrolithiasis are erroneously instructed to eliminate all calcium from their diet, a suggestion that can result not only in bone demineralization, particularly in women and children, but also in an increase in stone formation.[2-4,26] Sugar-sweetened soda appears to be associated with a greater risk for stone formation, whereas consumption of coffee, tea, beer, wine, and orange juice appears to be associated with a lower risk.[27] The exact mechanism of nephrolithiasis from soda is not known but is likely related to the high fructose content of soda, which may trigger a vasopressin-dependent concentration of urine, lower urinary pH, and cause uricosuria.[28,29]

***Physical examination.*** Most patients with idiopathic hypercalciuria are healthy and have normal findings on physical examination. Patients with hyperuricosuria and uric acid stone formation may display tophi. Central obesity is associated with a predisposition to metabolic syndrome and uric acid stones. Paraplegic patients with a chronic indwelling bladder catheter may be predisposed to chronic UTI and struvite stones.

***Laboratory findings.*** Urine pH is generally high in patients with struvite and calcium phosphate stones but low in patients with uric acid and calcium oxalate stones. Bacteriuria with urine pH greater than 7.0 suggests struvite stones. Urine should be cultured, and because many bacteria produce urease even when urine bacterial colony counts are low, the microbiology laboratory should be instructed to type the organism even if there are fewer than $10^5$ colony-forming units/mL. The specific gravity, if high, will confirm inadequate fluid intake in many patients. Hematuria may imply active stone disease with crystal or stone passage. Examination of the urine may reveal red blood cells along with characteristic crystals (Fig. 60.3).

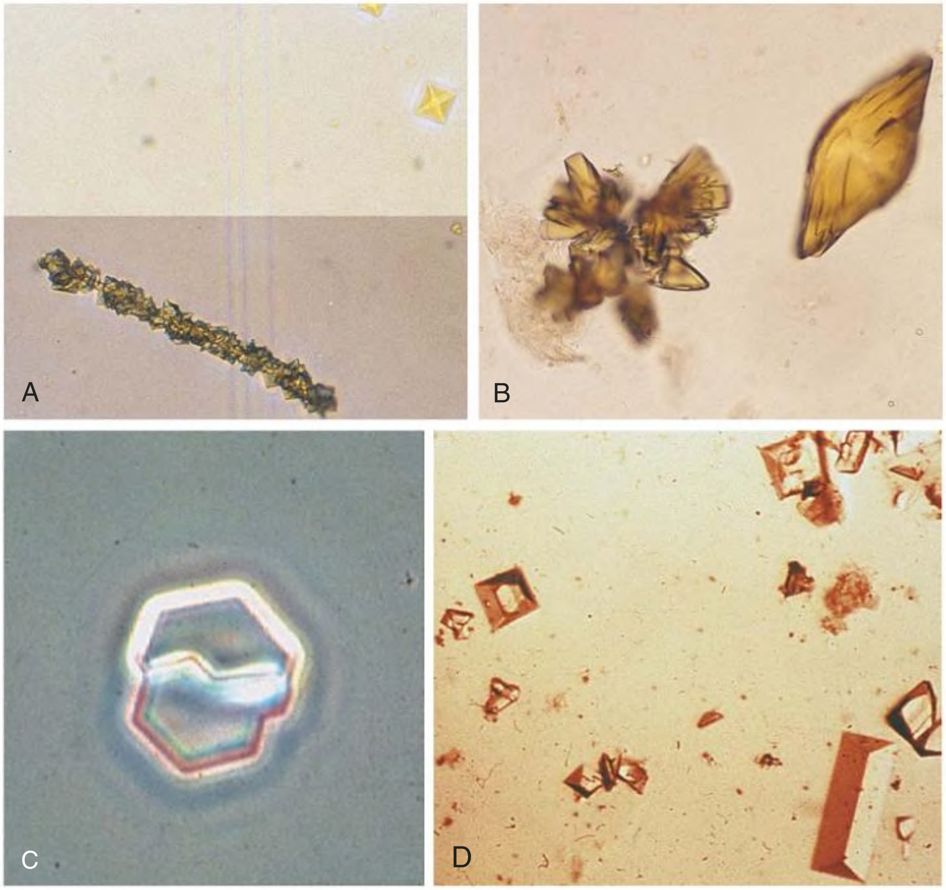


**Fig. 60.3 Urine Crystals.** (A) Oxalate crystals: a pseudocast of calcium oxalate crystals accompanied by crystals of calcium oxalate dihydrate. (B) Uric acid crystals: complex crystals suggestive of acute uric acid nephropathy or uric acid nephrolithiasis. (C) A typical hexagonal cystine crystal; a single crystal provides a definitive diagnosis of cystinuria. (D) Coffin lid crystals of magnesium ammonium phosphate (struvite). (Courtesy Dr. Patrick Fleet, University of Washington, Seattle.)

Blood tests required in the basic evaluation are serum electrolytes (sodium, potassium, chloride, and bicarbonate), creatinine, calcium, phosphorus, and uric acid. If the serum calcium is elevated or at the upper limit of normal, serum parathyroid hormone level should be measured, especially if the serum phosphorus is low. A low potassium or bicarbonate level may indicate a cause for hypocitraturia, such as distal renal tubular acidosis.

***Stone analysis.*** Patients should be encouraged to retrieve any stone they excrete for chemical analysis, which may help define the underlying metabolic abnormality and guide therapy.

***Imaging.*** Current imaging techniques used to evaluate stones include plain abdominal radiograph, unenhanced helical computed tomography (CT), and kidney ultrasound. Plain abdominal radiography performed with views of the kidneys, ureters, and bladder (KUB) may reveal opacifications in the areas of the kidneys and ureters that could be a result of calcium, cystine, or struvite stones (Fig. 60.4). Uric acid and xanthine calculi are radiolucent and will not be visible on plain films. Unenhanced helical CT scanning, also known as spiral CT, or CT urography, has replaced contrast intravenous urography (IVU) as a diagnostic test for acute ureteral colic because it is more sensitive and specific for ureteral stones and ureteral obstruction, and avoids the need for contrast (see Fig. 60.5A). Results from CT urography are available in minutes rather than hours, and this is an advantage in the emergency setting. There are several disadvantages of CT imaging: radiation dose from a CT urography is approximately three times that of a conventional IVU; CT urography is more expensive and is associated with a high rate of incidental findings that could lead to inappropriate referral and treatment. In addition, an experienced radiologist is required for optimal interpretation of the images, and may not be available at all times in urgent care facilities. CT urography should be avoided or limited in patients at risk for radiation exposure, such as children and pregnant women. Low-dose CT reduces radiation exposure and provides similar information to that provided by standard CT. However, low-dose CT is not recommended for patients with a body mass index greater than 30 kg/$m^2$.[30]

Kidney ultrasound has a lower sensitivity and specificity than CT but does not expose patients to ionizing radiation and is less expensive. A recent study indicated that ultrasonography may be a better choice than CT, especially for those patients who must avoid radiation exposure (Fig. 60.5B). In a multicenter, comparative effectiveness trial, 2759 patients who presented to the emergency department (ED) with suspected nephrolithiasis were randomized to undergo either initial ultrasound or abdominal CT.[31] Ultrasound was associated with lower cumulative radiation exposure than initial CT. There were no significant differences in high-risk diagnoses with complications (e.g., abdominal aortic aneurysm with rupture, pneumonia with sepsis), serious adverse events, pain scores, return visits to the emergency department, or hospitalizations between the ultrasound (whether assigned to either point of care in the emergency department or to the radiology department) and CT. Ultrasonography is the first-line imagining modality for pregnant women and patients younger than 14 years.[30]

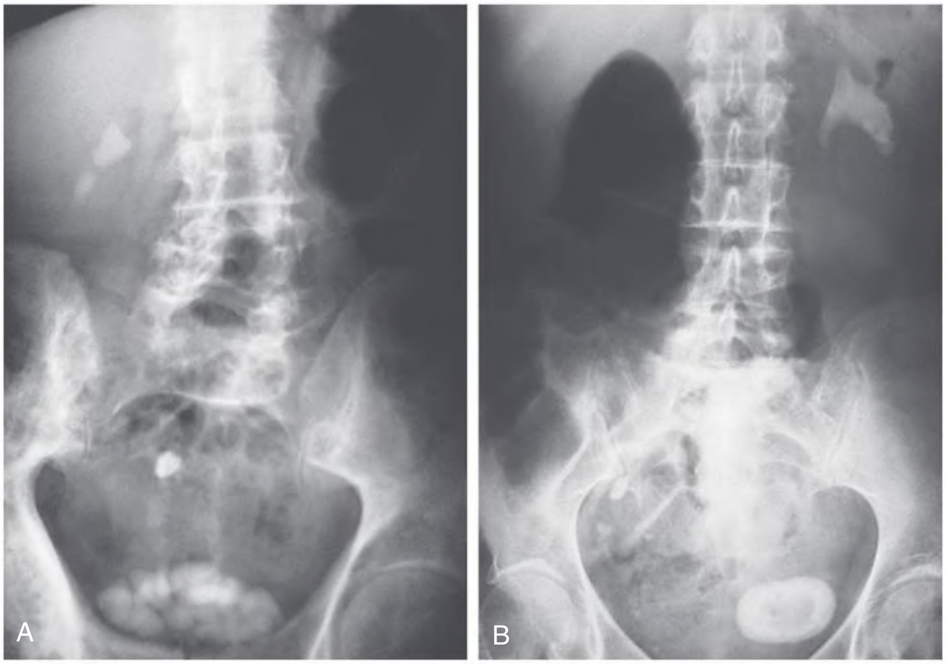


**Fig. 60.4** Radiopaque Kidney Calculi. (A) Radiograph showing multiple cystine stones in the right kidney, right ureter, and bladder. (B) Struvite stones: left staghorn calculus and a single bladder stone.

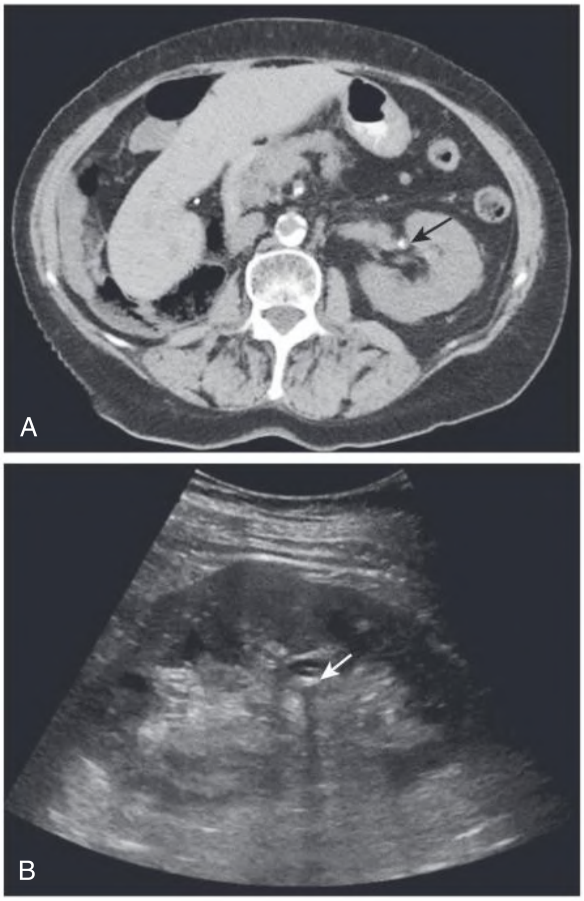


**Fig. 60.5** Kidney Calculi. (A) Unenhanced computed tomography scan showing a nonobstructing left kidney stone *(arrow)* in a patient with right nephrectomy. (B) Ultrasound of the left kidney (sagittal view) from the same patient showing a stone with an acoustic shadow behind it *(arrow)*.

Periodic monitoring, if deemed necessary, should be obtained with an ultrasound scan rather than KUB or CT, whenever possible, to minimize radiation exposure. The combination of ultrasound and KUB is more sensitive in detecting stones than either test alone, while minimizing radiation compared with CT. Many patients with stones have recurrent disease and may require repeated imaging. Children and young adults are at higher risk for consequences from radiation exposure, and studies in this population should be limited. Especially when caring for patients who may father or subsequently carry offspring, every effort should be made to limit radiation exposure. In general, we do not order routine follow-up radiographic examinations and do not order radiographic examinations unless patients are symptomatic and the results will alter subsequent care. If radiographs are ordered, every effort should be made to limit radiation exposure, as the results of a noncontrast study alone are often sufficient to guide therapy.

## Complete Evaluation

A complete evaluation should be arranged for patients with multiple or metabolically active stones (i.e., stones that increase in size or number within a year), in all children, in patients from demographic groups not typically prone to stone formation, and in those with stones other than those containing calcium.[23]

The complete evaluation should include a measure of urine volume and the quantity of calcium, oxalate, phosphorus, uric acid, sodium, citrate, and creatinine excreted in a 24-hour urine collection (Table 60.2). Urine creatinine is useful in assessing adequacy of the collection; men should excrete more than 15 mg/kg (132.6 μmol/kg), and women should excrete more than 10 mg/kg (88.4 μmol/kg) daily. Patients should be encouraged to perform the urine collection on a typical day while eating their usual diet, although many patients prefer to collect the urine on weekends when their diet and habits may differ from those during usual workdays. Specialized testing, such as urine collections during high- or low-dietary calcium intake, is not recommended. Careful instructions should be given to avoid overcollection or undercollection. Patients should be instructed to discard the first morning

TABLE 60.2 **Optimal 24-Hour Urine Values in Recurrent Nephrolithiasis**

| | |
|---|---|
| **24-Hour Urine Values** | |
| Volume | >2–2.5 L |
| Calcium | <4 mg/kg (0.1 mmol/kg), ~300 mg (7.5 mmol) in men, ~250 mg (6.3 mmol) in women |
| Oxalate | <40 mg (0.36 mmol) |
| Uric acid | <750 mg (4.5 mmol) in women and <800 mg (4.7 mmol) in men (can be pH dependent) |
| Citrate | >320 mg (17 mmol) |
| Sodium | <2000 mg (87 mmol) |
| Phosphorus | <1100 mg (35 mmol) |
| Creatinine | >10 mg/kg (88 µmol/kg) in women and >15 mg/kg (132 µmol/kg) in men, if specimen is a complete collection |
| **Urine Supersaturation Values** | |
| Calcium oxalate supersaturation | <5 |
| Calcium phosphate supersaturation | 0.5–2 |
| Uric acid supersaturation | 0–1 |

urine at the start of the collection and collect all urine for the next 24 hours, including the first urine collection on the second morning.

A disadvantage of the standard 24-hour urine collection is that laboratories vary in the preservatives required to process the various constituents. Many laboratories require more than one collection to measure all the urinary constituents, reducing compliance and therefore the accuracy of the results, and supersaturation is often not calculated.

A better approach available in some speciality laboratories is to undertake a 24-hour urine collection for the relevant urinary constituents and calculation of supersaturation for the common solid phases of calcium oxalate, brushite (calcium phosphate), and uric acid (see Table 60.2). Urine for supersaturation analysis has been shown to correlate well with stone composition, and the risk for stone formation rises with increasing supersaturation. Determination of supersaturation is far more informative and helpful for planning proper therapy than evaluation of only the individual urinary constituents.

Patients can bring their specimens to a local laboratory or may send them to specialized laboratories that measure calcium, oxalate, citrate, uric acid, creatinine, sodium, potassium, magnesium, sulfate, phosphorus, chloride, urine urea nitrogen, and pH. Calculation of supersaturation from a 24-hour urine specimen will provide values lower than the peak postprandial supersaturation and peak nighttime supersaturation, either of which may initiate stone formation. Patients should stop taking any vitamin C or multivitamins containing more than 100 mg of vitamin C for at least 5 days before the urine collection because the antioxidants in the vitamin may interfere with the assay.

## General Treatment

Intervention for stone removal may be required when pain, obstruction, and/or infection resulting from nephrolithiasis do not respond to conservative management. Surgical management of stones includes extracorporeal shock wave lithotripsy (ESWL) and both endoscopic and percutaneous surgical removal of stones, and is further discussed in Chapter 63. The risk for developing reduced glomerular filtration rate (GFR) varies with the type of stone, and this must be considered in planning management.

## Medical Management

Patients who are seen by stone "specialists" often have a substantial decrease in stone recurrence even without pharmacologic intervention.[32] This phenomenon, termed the *stone clinic effect,* is likely a result of modifications in diet and fluid intake. These nonpharmacologic measures include an increase in fluid intake, which increases urine volume; restriction of dietary sodium, which leads to a reduction of urine calcium excretion; restriction of animal protein, which also leads to a reduction of urine calcium and uric acid excretion and an increase in excretion of the calcification inhibitor citrate; and ingestion of an age- and sex-appropriate amount of dietary calcium, which will bind oxalate in the intestine, preventing its absorption. Although dietary calcium restriction continues to be prescribed by many physicians, substantial evidence indicates that this is not beneficial and can actually increase the rate of stone formation (see later discussion of calcium stones) and can promote bone demineralization.[4,26]

***Fluid intake.*** An increase in urine volume to more than 2 to 2.5 L daily has been proven to reduce the incidence of stones. Large urine volumes will reduce calcium oxalate supersaturation, as well as precipitation of other crystals. Increased fluid intake to augment urine volume is also a mainstay of therapy for patients with uric acid and cystine stones. The period of maximum risk for stone formation is at night, when urine concentration is physiologically increased. Patients should be encouraged to drink enough fluid in the evening to provoke nocturia and then drink further fluid before returning to bed. Patients should avoid calorie-containing beverages, such as sugar-sweetened beverages, which may lead to weight gain and are associated with a higher risk of stone formation.[27]

***Salt intake.*** Urine sodium excretion correlated directly with urine calcium excretion;[2,3] thus, dietary salt restriction is associated with decreased urine calcium excretion. Patients should be instructed to limit daily sodium intake to a maximum of 2 g (87 mmol).

***Dietary protein.*** Animal protein ingestion increases the frequency of kidney stone formation by a number of mechanisms. Metabolism of certain amino acids leads to generation of sulfate ions, which render urinary calcium ions less soluble.[33,34] The metabolic acidosis that results from excess protein ingestion causes calcium release from bone and a consequent increase in the filtered load of calcium.[33,34] Acidosis also decreases tubular calcium reabsorption, resulting in hypercalciuria. Urinary citrate excretion is also pH dependent, with acidosis leading to a decrease in citrate excretion. The result of increased animal protein intake is an increase in urinary calcium excretion that is rendered less soluble because of concomitant sulfate excretion and hypocitraturia. Low urine pH, coupled with increased uric acid excretion from the metabolism of animal protein, can result in uric acid lithiasis. Stone formers should be encouraged to consume a moderate protein intake. Dietary fructose may also increase uric acid lithiasis.

***Dietary calcium.*** Despite conventional wisdom, several studies have demonstrated a decrease in stone incidence when people consume diets adequate, but not excessive, in calcium (>1500 mg/day of calcium).[35] This beneficial effect has been attributed to intestinal binding of ingested oxalate (which is highly lithogenic) by dietary calcium preventing oxalate absorption. Although women have reduced stone formation while ingesting an age-appropriate amount of dietary calcium, this benefit may not extend to those taking calcium in the form of calcium supplements.[36] The data are inconsistent as to whether calcium supplements increase the risk for nephrolithiasis. Some have postulated that any increased risk may be a result of timing the supplemental calcium ingestion apart from meals, which would enhance calcium absorption without reducing oxalate absorption. In addition, the calcium supplements may dissociate rapidly, leading to rapid absorption, increased filtered calcium load, and transitory hypercalciuria, leading to increased supersaturation.

We advise patients that it is best to obtain calcium from dairy products rather than supplements.

Women ingesting calcium supplements may be at increased risk for cardiovascular disease and death compared with those not taking supplements, perhaps by promoting vascular calcification.[37] In one long-term study of more than 60,000 Swedish women, those with a dietary calcium intake exceeding 1400 mg/day had an increased risk for cardiovascular mortality with an HR of 1.49 (95% CI, 1.09–2.02). The risk for all-cause mortality rose to 2.57 (95% CI, 1.19–5.55) when any calcium tablet supplements were added to this high dietary intake.[38] A review of cardiovascular mortality in over 15 studies involving calcium supplementation versus placebo showed increased risk for myocardial infarction in those randomized to calcium supplements with an HR of 1.31 (95% CI, 1.02–1.67), which rose to 1.85 (95% CI, 1.28–2.67) in patients already taking more than 805 mg of dietary calcium daily.[37] In a 10-year follow-up of 5448 adults free of clinically diagnosed cardiovascular disease, calcium supplements increased the risk for incident coronary artery calcification (relative risk 1.22 [95% CI, 1.07–1.39]).[39] The Institute of Medicine in the United States recently adjusted the recommended allowance of calcium to 1000 mg/day for adults older than 19 years and 1200 mg/day for women older than 50 years.[40] We recommend that women take the appropriate amount of calcium in the form of food with limited supplement intake, except in cases of severe osteoporosis with insufficient dietary intake.

Support for the use of an age- and sex-appropriate amount of dietary calcium was provided by a randomized prospective study comparing the rate of stone formation in men assigned a low-calcium diet with those assigned a normal-calcium, low-sodium, and low–animal protein diet.[26] The men assigned the low-calcium diet were twice as likely to have recurrent stones over 5 years compared with those on the normal-calcium, low-sodium, and low–animal protein diet. Urinary calcium oxalate supersaturation also diminished more rapidly in those on the higher-calcium diet and remained lower than that of men on the low-calcium diet for most of the 5-year study. This reduction in supersaturation was the result of a greater fall in urinary oxalate in the men eating the normal-calcium, low-sodium, and low–animal protein diet.[26,41]

Patients prescribed a low-calcium diet can avoid excessive hyperoxaluria when adequately instructed to also consume a low-oxalate diet.[42] Some contend that this approach may benefit patients with excessive intestinal absorption of calcium associated with severe hypercalciuria by allowing calcium restriction without the risk for significant osteopenia. We recommend, however, an age- and sex-appropriate calcium intake, best derived from a diet containing an adequate number of dairy products. Because stone formation can be reduced with normal calcium intake and there is a risk for bone demineralization and osteoporosis with calcium restriction, we consider the low-calcium diet to be obsolete.[24,38]

***Vitamin D.*** Given the great degree of vitamin D deficiency and insufficiency in northern latitudes, vitamin D supplementation is very common.[43] There is concern that this might exacerbate nephrolithiasis, given the role of vitamin D in mineral metabolism. Active forms of vitamin D, such as calcitriol, may increase urinary calcium excretion.[44] However, studies have shown no association between vitamin D supplementation in the form of ergocalciferol, cholecalciferol, or levels of serum 25-hydroxyvitamin D and calcium excretion, nor an increased rate of stone formation in hypercalciuric patients on vitamin D supplementation. Thus, vitamin D therapy should not be withheld in patients with vitamin D deficiency (25-hydroxyvitamin D levels <20 ng/mL) on the basis of kidney stone disease.[45] If needed, we prefer cholecalciferol ($D_3$) to ergocalciferol ($D_2$) because of its more robust biologic activity and longer half-life.

## SPECIFIC TYPES OF STONES

### Calcium Stones

Calcium-containing kidney stones constitute approximately 70% of all stones formed. Most calcium stones are composed of calcium oxalate, either alone or in combination with calcium phosphate or uric acid. A small percentage of stones are composed entirely of calcium phosphate.[2] Most calcium stones do not exceed 1 to 2 cm in width. Surgical intervention is often required for stones larger than 5 mm.

Calcium-based stones most often develop as a result of hypercalciuria. Other causes of calcium stones are hyperoxaluria, hyperuricosuria, hypocitraturia, renal tubular acidosis, certain medications, and congenital abnormalities of the genitourinary tract (Fig. 60.6). Specific therapy for patients with calcium stones depends on the underlying metabolic abnormalities detected on evaluation. General therapy as outlined earlier always should be instituted; however, more definitive treatment is often required to significantly decrease the rate of recurrent stone formation.

#### Hypercalciuria

***Etiology.*** Hypercalciuria for which a causative metabolic abnormality cannot be determined is called *idiopathic hypercalciuria.* Previously, hypercalciuric patients were divided into those with excessive kidney calcium excretion and those who absorbed excessive amounts of calcium via the gastrointestinal tract. However, hypercalciuric patients generally do not have a transport defect limited to a single site; they appear to have a systemic dysregulation of calcium transport at the major calcium transporting sites, the intestine, kidney, and bone. In both hypercalciuric rats and in humans placed on a low-calcium diet, there is a wide range of calcium excretion. Hypercalciuric patients may appear to have excessive excretion on one examination and excessive absorption on another.[2,3] Some patients excrete more calcium than they consume, indicating a negative total body calcium balance. This calcium must be derived from the mineral phases of bone, which contain by far the largest repository of calcium in the body. The cause of this systemic disorder in calcium transport has, in hypercalciuric stone–forming rats and perhaps in humans, been linked to an increased number of biologically active receptors for vitamin D in the major calcium transporting organs (intestine, kidneys, and bone).[46,47] Metabolic disorders leading to elevated levels of serum calcium, parathyroid hormone, or $1,25(OH)_2D$ (1,25-dihydroxyvitamin D) also may result in hypercalciuria.

***Treatment.*** For hypercalciuria the first-line therapy is a thiazide diuretic, which acts to reduce urinary calcium. In the United States, chlorthalidone 25 to 50 mg is the drug of choice because it requires only once-daily administration. Indapamide 1.25 to 2.5 mg/day does not tend to raise serum lipids as much as other thiazides and may be preferred for patients with cardiac risk factors or elevated serum lipids. On commencing these medications, patients should be instructed to increase their dietary potassium intake, and a serum potassium level should be checked 7 to 10 days later. If serum potassium is low, oral potassium supplementation should be initiated. Potassium citrate is generally preferred over potassium chloride. However, most patients find potassium citrate liquid preparations unpalatable. A wax matrix tablet of potassium citrate is well tolerated and is available in some countries. In general, patients are able to maintain a normal serum potassium level with potassium citrate 20 to 40 mmol/day. The serum potassium and bicarbonate levels should be rechecked 7 to 10 days later for further dosage adjustment. Because citrate is a base, potassium citrate may excessively raise the serum bicarbonate level and urinary pH, which could promote calcium phosphate stone formation. Urinary pH and supersaturation must be carefully monitored, and a

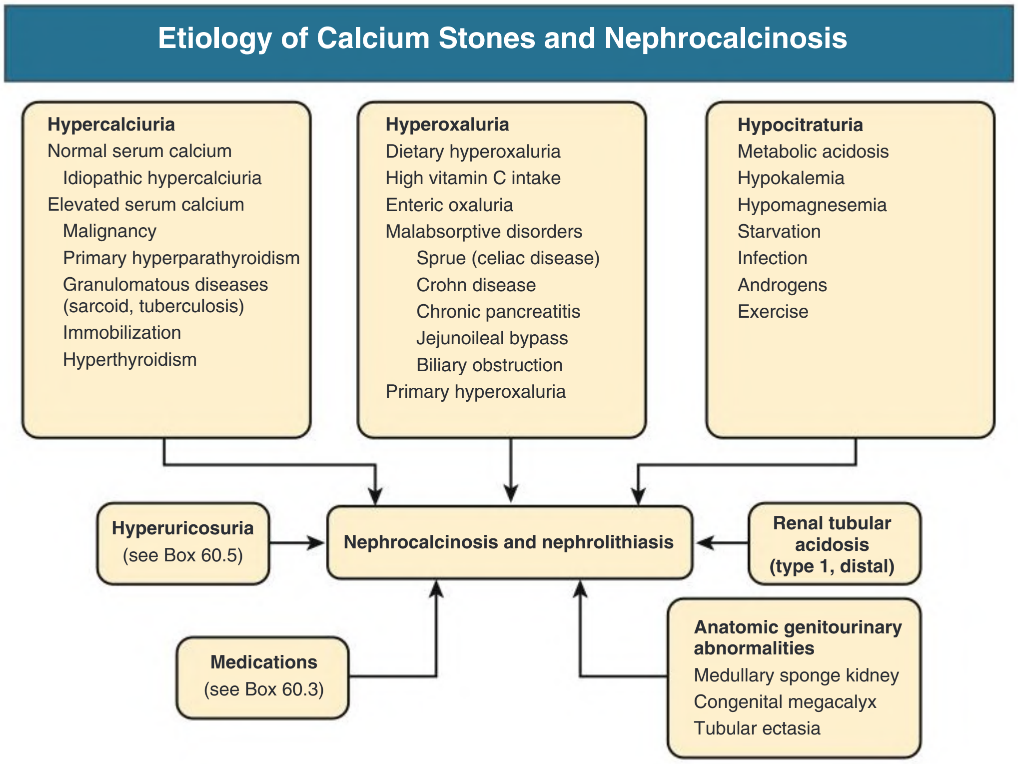


**Fig. 60.6** Etiology of calcium stones and nephrocalcinosis.

change to potassium chloride may be required if supersaturation of calcium phosphate worsens with increasing urinary pH in a patient with primarily calcium phosphate stones.

The 24-hour urine calcium, sodium, and citrate should be rechecked after several weeks to months. If the calcium excretion remains elevated, the thiazide dose should be increased. However, if the sodium excretion also remains high (i.e., >100 mEq/day), patients should be encouraged to limit their sodium intake further because they will not have an adequate response to the diuretic on a high-sodium diet. If serum potassium remains low despite supplementation or the calcium excretion remains high despite increased thiazide dosing, addition of a potassium-sparing diuretic may be required to increase serum potassium. Amiloride with a starting dose 5 mg/day is the preferred treatment for thiazide-induced hypokalemia.[2,3] Triamterene should not be used because it can precipitate into stones.

***Dietary recommendations.*** See the discussion of general treatment.

## Hyperoxaluria

***Etiology.*** Elevated urinary oxalate results from excessive dietary intake (dietary oxaluria), gastrointestinal disorders that can lead to malabsorption (enteric oxaluria), or an inherited enzyme deficiency that results in excessive metabolism of oxalate (primary hyperoxaluria) (see Fig. 60.6).[48]

Dietary excess of oxalate generally does not raise urinary oxalate above 60 mg/24 h (0.54 mmol/24 h). Enteric oxaluria may occur when malabsorption results in excessive colonic absorption of oxalate, as a result of sprue (celiac disease), Crohn disease, chronic pancreatitis, or short bowel syndrome or after bariatric surgery. The anion exchange transporter Slc26a6 appears responsible for intestinal oxalate secretion, and mice with targeted inactivation of this transporter have hyperoxaluria.[49] It is not yet known whether patients with enteric hyperoxaluria have mutations in Slc26a6. Urinary oxalate is generally more than 60 mg/24 h (0.54 mmol/24 h) and can exceed 100 mg/24 h (0.9 mmol/24 h). In primary hyperoxaluria, the tremendous oxalate production results in widespread calcium oxalate deposition throughout the body at an early age. This infiltration of calcium oxalate into organs can result in cardiomyopathy, bone marrow suppression, and kidney failure. Urinary oxalate values may range from 80 to 300 mg/24 h (0.72–2.70 mmol/24 h).

There are three types of primary hyperoxaluria with unique enzymatic defects in the liver glyoxylate pathway. In type 1, the defective enzyme is alanine–glyoxylate aminotransferase, encoded by the gene *AGXT* on chromosome 2; this accounts for approximately 80% of cases. Type 2 tends to be milder and is caused by mutations in the *GRHPR* gene on chromosome 9, which results in failure of glyoxylate reduction to glycolate; this accounts for approximately 10% of the cases. Type 3 is a result of mutations in the gene that encodes the mitochondrial 4-hydroxy-2-oxoglutarate aldolase enzyme, which catalyzes the cleavage of 4-hydroxy-2-oxoglutarate to pyruvate and glyoxylate and accounts for about half of the remaining cases.[50]

***Treatment of dietary and enteric hyperoxaluria.*** Treatment of dietary oxaluria consists of dietary oxalate restriction. Patients should be given a list of foods that have high oxalate content to avoid or eat in moderation (see Box 60.4). Calcium-containing food may be included at each meal to bind intestinal oxalate and prevent its absorption. To evaluate the success of dietary modification, 24-hour urine oxalate and supersaturation of calcium oxalate can be used.

Specific therapy for the malabsorptive disorder, such as a gluten-free diet for patients with sprue, is the first-line treatment for enteric hyperoxaluria. More generalized therapy for steatorrhea, such as a low-fat diet, cholestyramine, and administration of medium-chain triglycerides, may reduce fat malabsorption as well as oxalate absorption and subsequent excretion. The low-oxalate diet and mealtime calcium carbonate prescribed for patients with dietary oxaluria is also helpful for these patients. The diarrhea associated with these disorders may result in low urine volumes, hypokalemia, hypocitraturia, and hypomagnesuria. Patients should therefore be advised to increase their fluid intake and take potassium citrate (in this case, the liquid, although unpalatable, is better absorbed than the tablets), as well as a magnesium supplement. Magnesium also serves as a urinary stone inhibitor and can be given as magnesium gluconate 0.5 to 1 g once or twice a day or as magnesium oxide 400 mg every 12 hours.

***Treatment of primary hyperoxaluria.*** Primary hyperoxaluria type 1 (PH1) is a severe disorder that until recently could be cured only with liver transplantation to replace the defective hepatic enzyme. Traditional treatments for PH1 have included pyridoxine (vitamin $B_6$) in doses ranging from 2.5 to 15 mg/kg/24 h in an attempt to reduce oxalate production. Efforts should be made to render the calcium and oxalate more soluble in the urine by raising the urinary pH (to at least 6.5) and giving supplemental citrate and magnesium. Orthophosphate is also an effective inhibitor of urinary calcium oxalate precipitation and can be safely administered in patients with a GFR greater than 50 mL/min/1.73 $m^2$. Because oxalate is poorly excreted in CKD and is not removed well by dialysis, kidney transplantation serves not only to improve kidney function but also to improve oxalate excretion and diminish systemic oxalosis.[16] Recently, a specific RNA interference therapeutic, lumasiran, has been approved in Europe and the United States for treatment of PH1. This therapy leads to decreased synthesis of glycolate oxidase, an enzyme upstream of the defect in PH1. Lumasiran leads to a prolonged and substantial reduction in urinary oxalate excretion and plasma oxalate concentration, both of which generally decrease to near normal levels in both children and adults.[51] Lumasiran promises to be the standard of care for all patients with PH1. Hopefully with this important therapeutic advance, the devastating clinical consequences of PH1 will be avoided, including kidney failure and systemic oxalosis. Similar strategies to treat PH2 and PH3 are being developed.

*Oxalobacter formigenes* primarily uses oxalate for cellular metabolism.[52] Calcium oxalate stone formers who are colonized with *O. formigenes* have lower urine oxalate levels than those who are not colonized.[53] A small clinical trial involving administration of *O. formigenes* resulted in a modest decrease in urinary oxalate in some patients.[52] Further studies with larger numbers of patients and assessing clinically relevant outcomes are required.

### Hypocitraturia

Citrate inhibits stone formation. A number of conditions reduce urinary citrate excretion, predisposing to stone formation. Excessive protein intake, hypokalemia, metabolic acidosis, exercise, hypomagnesemia, infections, androgens, starvation, and acetazolamide have all been implicated in decreased urinary citrate excretion. Therapy involves treatment of the underlying condition and potassium citrate supplementation. The potassium salt is preferred to sodium citrate because sodium promotes kidney calcium excretion. Potassium citrate 15 to 25 mmol two to three times daily is required, and tablets are considered by most patients to be more palatable than the liquid preparation. In patients with reduced GFR, serum potassium should be monitored carefully, and dose reduction may be needed if hyperkalemia develops.[16]

### Distal Renal Tubular Acidosis

Patients with distal (type 1) renal tubular acidosis have impaired distal tubular excretion of hydrogen ions with non–anion gap metabolic acidosis and alkaline urine. The acidosis causes calcium and phosphate to be released from bone with an ensuing increase in kidney excretion of these ions.[33,34] The acidosis also leads to an increase in citrate reabsorption by the proximal tubule. The end result is a high urinary pH, hypocitraturia (urinary citrate generally <100 mg/24 h [0.53 mmol/24 h]), and increased kidney excretion of calcium and phosphate, all of which increase the propensity for calcium phosphate precipitation. Nephrocalcinosis in this setting is not uncommon, because calcium precipitates with phosphorus in the alkaline tubular fluid. The metabolic acidosis and hypocitraturia should be treated with a combination of sodium citrate (or bicarbonate) and potassium citrate (or bicarbonate). Often large amounts (e.g., 1–2 mmol/kg/day in 2 or 3 divided doses) are required to neutralize the acidosis.[2] Citrate is generally preferred to bicarbonate by patients because it does not produce carbon dioxide on contact with stomach acid with resultant gastrointestinal bloating. However, citrate is usually more expensive than bicarbonate and promotes the absorption of aluminum.

### Hyperuricosuria

Calcium oxalate crystals often nucleate around other crystal types such as uric acid. Hyperuricosuria contributes to nephrolithiasis in 10% to 15% of calcium stones. Patients with hyperuricosuric calcium oxalate nephrolithiasis may have hyperuricosuria with normal urinary calcium and oxalate, but often a higher urinary pH (>5.5) than patients with pure uric acid stones. Therapy for hyperuricosuria consists of increased fluid intake and reduced dietary purine intake. If uric acid excretion remains elevated, allopurinol should be initiated at 100 to 300 mg/day.[2]

## Uric Acid Stones

### Epidemiology

The prevalence of uric acid stones depends greatly on geographic location. In the United States, uric acid stones constitute 5% to 10% of all stones formed, whereas in Mediterranean and Middle Eastern countries, uric acid stones may constitute up to 75% of stones. The stones are radiolucent and therefore poorly visible on plain radiographs. They are detectable on ultrasound and CT and as filling defects on IVU (Fig. 60.7).

### Etiology and Pathogenesis

Causes of hyperuricosuria include excessive dietary purine or protein intake, disorders associated with cellular breakdown (tumor lysis syndrome, myeloproliferative disorders, hemolytic anemia), gout, uricosuric medications, certain inborn errors of metabolism, and possibly excessive fructose intake.[54]

Three major factors influence uric acid stone formation: low urine pH, low urine volume, and elevated urinary uric acid levels (Box 60.5). Of the three, low urine pH is the principal metabolic disorder found in patients with uric acid nephrolithiasis. Uric acid is poorly soluble at pH below 5.5. Solubility increases with urine alkalinity such that at urine pH 6.5, urine can contain over 6 times the quantity of uric acid present at pH 5.3 without exceeding supersaturation. The rising incidence of obesity and insulin resistance in the United States led to a parallel increase in uric acid lithiasis. The urinary acidosis is likely a result of impaired ammoniagenesis, which results in excessive excretion of unbuffered acid and a very low urine pH.[9,55] The geographic and ethnic variations in the incidence of uric acid stones may be due to diet, caused by environmental factors, or the result of genetic susceptibility in some populations.[9]

### Treatment

Treatment of uric acid stones involves increasing urine pH and volume, and decreasing uric acid excretion. Alkaline urine not only can prevent uric acid stone formation but also may result in stone dissolution. To raise urine pH, potassium citrate is recommended. Although sodium bicarbonate alkalinizes the urine and enhances uric acid solubility, the added sodium increases sodium urate formation, which

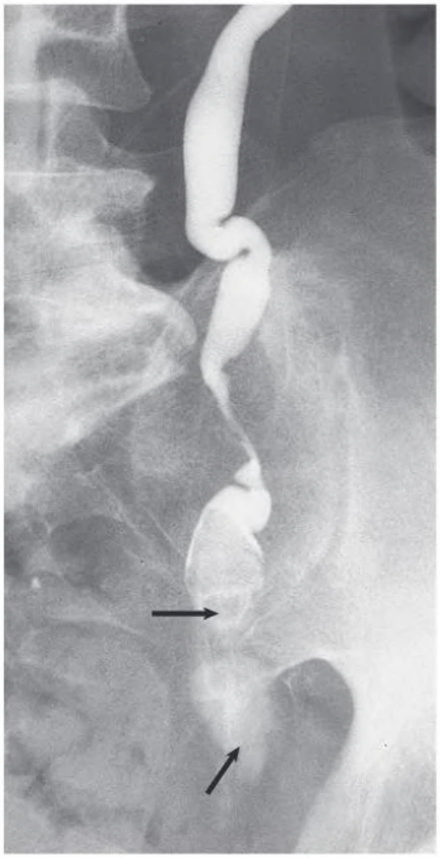

**Fig. 60.7 Radiolucent Urate Calculi.** Antegrade pyelogram showing multiple radiolucent uric acid stones *(arrows)* obstructing the lower ureter.

## BOX 60.5 Factors Associated With Uric Acid Stone Formation

**Low Urine pH (≤5.5)**
- High–animal protein diet
- Diarrhea
- Insulin resistance (high body mass index, metabolic syndrome, type 2 diabetes)

**Low Urine Volume**
- Inadequate fluid intake
- Excessive extrarenal fluid losses
  - Diarrhea
  - Insensible losses (e.g., perspiration)

**Hyperuricosuria**
- Excessive dietary purine intake
- Hyperuricemia
  - Gout
  - Intracellular-to-extracellular uric acid shift
  - Myeloproliferative disorders
  - Tumor lysis syndrome
- Inborn errors of metabolism
  - Lesch-Nyhan syndrome
  - Glucose-6-phosphatase deficiency

**Medications**
- See Box 60.3

serves as a nidus for calcium oxalate precipitation. Potassium citrate 40 to 50 mmol/day in divided doses is given, increasing the dose as necessary to achieve a urine pH of 6.5 to 7.0. Patients should monitor pH with urine dipsticks at various times of the day and adjust the dosage accordingly. If urine pH remains low despite potassium citrate exceeding 100 mmol/day, or if that dose results in hyperkalemia, acetazolamide may be added. This carbonic anhydrase inhibitor produces an alkaline urine similar to that seen in renal tubular acidosis. Patients should be cautioned not to exceed urine a pH of 7.0 because this may result in calcium phosphate precipitation.

A low-purine and low–animal protein diet is also useful in raising urinary pH and decreasing uric acid excretion (see Box 60.4). If uric acid excretion remains high despite dietary intervention, as in patients with disorders of cellular catabolism, allopurinol should be prescribed, 100 mg/day increasing to 300 mg/day as needed to keep urinary uric acid excretion lower than 750 mg/24 h (4.5 mmol/24 h). See Box 60.4 for high-purine foods to avoid.

## Struvite Stones

Struvite stones are also referred to as *infection stones* or *triple phosphate stones.* The stones grow rapidly to a large size, can reduce GFR in the affected kidney, and are difficult to eradicate. Because of the significant morbidity in patients with struvite stones, they also have been termed *stone cancer*. Most staghorn calculi, large stones that penetrate more than one renal calyx, are composed of struvite. Their formation requires the presence of urease-producing bacteria in the urine (Box 60.6).[56]

### Etiology and Pathogenesis

Struvite stones are composed of magnesium ammonium phosphate and calcium carbonate apatite. They form when urease production by certain urinary bacteria breaks down urea to ammonium ($NH_4^+$) and a carboxyl ($OH^-$) group. The urine becomes quite alkaline; urinary phosphate becomes insoluble and forms a solid phase with magnesium, calcium, and the ammonium. Women are more prone to struvite nephrolithiasis than men because of an increased propensity for UTI. Others predisposed to developing struvite stones through infections or urinary stasis include patients with indwelling urinary catheters, neurogenic bladders, genitourinary tract anomalies, and spinal cord lesions. An alkaline urine (pH ≥7.0), urine culture of urease-producing bacteria, and large stones suggest the diagnosis of struvite nephrolithiasis.[56]

Certain gram-negative and gram-positive bacteria have been implicated in urease production and consequent struvite formation, with the most common being *Proteus* spp. (see Box 60.6). *Escherichia coli,* which is frequently present in urine cultures, is not a urease producer. If there is a strong suspicion for struvite stones but no organism is

## BOX 60.6 Factors Associated With Struvite Stone Formation[a]

- Urease-producing bacteria
  - *Proteus*
  - *Haemophilus*
  - *Yersinia* spp.
  - *Staphylococcus epidermidis*
  - *Pseudomonas*
  - *Klebsiella*
  - *Serratia*
  - *Citrobacter*
  - *Ureaplasma*
- Elevated urinary pH

[a] Escherichia coli is not a urease producer.

detected in the urine, a specific culture request for *Ureaplasma urealyticum* should be considered, because it does not grow on routine culture media.

### Treatment

Struvite stones require aggressive medical and surgical management. Antibiotic therapy is important to reduce further stone growth and for stone prevention. Bacteria will remain in the stone interstices, however, and stones will continue to grow unless chronic antibiotic suppression is maintained or the calculi are completely eradicated. Given the need for complete stone removal to effect a cure, early urologic intervention is advised. Stones smaller than 2 cm may respond well to ESWL; however, larger stones will likely require percutaneous nephrolithotomy, often in combination with ESWL (see Chapter 63). Any stone fragments retrieved should be cultured and culture-specific antibiotics continued.[57] Once the urine is sterile, usually approximately 2 weeks after initiation of therapy, the dose is halved. Monthly urine cultures should be obtained, and if they remain sterile for 3 consecutive months, antibiotics may be discontinued, although surveillance urine cultures should continue monthly for a full year.

## Cystine Stones

Cystinuria is an autosomal disorder in which there is a tubular defect in dibasic amino acid transport, resulting in increased cystine, ornithine, lysine, and arginine excretion. It is also discussed in Chapter 50. The pattern of inheritance may be autosomal recessive or autosomal dominant with incomplete penetrance. The stone disease is usually clinically manifested by the second and third decades of life. Because of the high sulfur content of the cystine molecule, the stones are apparent on plain radiographs (see Fig. 60.4A) and often will manifest as staghorn calculi or multiple bilateral stones.

Cystine is poorly soluble, only approximately 300 mg/L (1.25 mmol/L) at a neutral pH. Normal cystine excretion of approximately 30 to 50 mg (0.12–0.21 mmol) per day is readily soluble in the usual daily urine output of approximately 1 L. However, homozygote cystinurics often excrete 250 to 1000 mg (1.04–4.20 mmol) of cystine per day, with heterozygotes excreting an intermediate amount. Treatment is directed at decreasing the urinary cystine concentration below the limits of solubility. Because the dietary precursor of cystine, methionine, is an essential amino acid, it is impractical to significantly reduce intake. Increasing urine volume so that cystine remains below the limits of solubility sometimes requires 4 L of urine per day. A low-sodium diet may reduce urine cystine excretion, although the mechanism by which this occurs is not clear.[58] Increasing urine pH greater than 7.5 will increase cystine solubility, but this is difficult to achieve on a long-term basis. Tiopronin 800 mg/day in 3 divided doses or D-penicillamine with a starting dose 250 mg/day and maximum dose of 2 g/day will both bind cystine and reduce urinary supersaturation; however, side effects may limit their use. The angiotensin-converting enzyme inhibitor captopril may be effective by forming a thiol-cysteine disulfide bond that is more soluble than cystine.[59]

## Stones Associated With Melamine Exposure

By late 2008, more than 50,000 Chinese children aged 3 years or younger had developed kidney stones associated with contaminated milk products. Powdered milk and infant formulas were noted to contain melamine, a nitrogenous substance synthesized from urea that increases the apparent protein content of the product.[22] Risk factors for nephrolithiasis after melamine exposure may include volume depletion, small body size, uricosuria, and low urine pH.

Although affected children often presented with dysuria and hematuria, many children who were subsequently tested were asymptomatic despite kidney stones identified on ultrasound.[22] On urinalysis, some children exhibited fan-shaped crystals. The kidney stones formed were radiolucent and fragile. Many were composed of a combination of uric acid with melamine and were amenable to dissolution by hydration and alkalinization.

**BOX 60.7 Causes of Nephrocalcinosis**

**Medullary**

***Disturbed Calcium Metabolism***
- Hyperparathyroidism
- Sarcoidosis
- Milk-alkali syndrome
- Rapidly progressive osteoporosis
- Idiopathic hypercalciuria

***Other Tubular Disease***
- Distal (type 1) renal tubular acidosis
- Oxalosis[a]
- Dent disease (X-linked hypercalciuric nephrolithiasis)
- X-linked hypophosphatemic rickets
- Bartter syndrome
- Hypomagnesemia-hypercalciuria syndrome

***Anatomic Disease***
- Medullary sponge kidney
  - Papillary necrosis

***Medications***
- Acetazolamide
- Amphotericin
- Triamterene

**Cortical**
- Cortical necrosis
- Transplant rejection
- Chronic glomerulonephritis
- Trauma
- Tuberculosis
- Oxalosis[a]

[a] Oxalosis typically causes both cortical and medullary nephrocalcinosis.

# NEPHROCALCINOSIS

Nephrocalcinosis refers to increased calcium content within the kidney and may be unilateral or bilateral.

## Etiology and Pathogenesis

### Medullary Nephrocalcinosis

Medullary nephrocalcinosis in which the calcification tends to occur in the area of the renal pyramids accounts for most cases of nephrocalcinosis. It is typically associated with elevated urinary calcium, phosphate, and oxalate, or it can occur with alkaline urine (Box 60.7). Any disorder that can lead to hypercalcemia and/or hypercalciuria may be implicated. Instead of stone formation, smaller parenchymal calcifications deposit in the medulla, and they are usually bilateral and relatively symmetric (Fig. 60.8). Some metabolic disorders, particularly oxalosis caused by primary hyperoxaluria, can result in both medullary and cortical nephrocalcinosis (Fig. 60.9).

In adults, the most common causes of medullary nephrocalcinosis are primary hyperparathyroidism, distal renal tubular acidosis, and

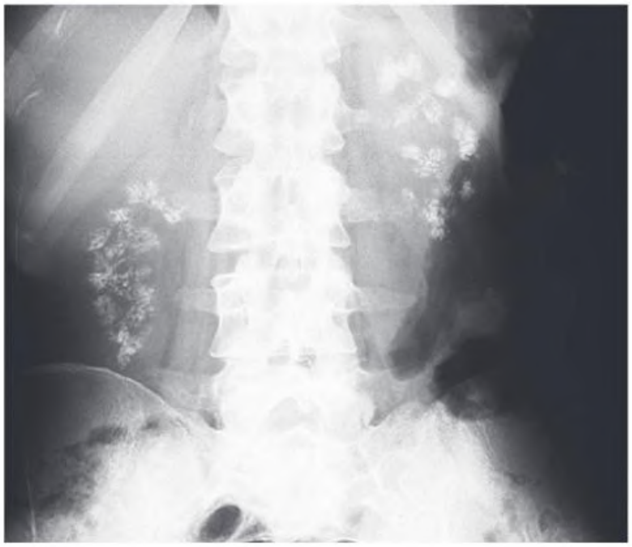

**Fig. 60.8** **Medullary Nephrocalcinosis.** Plain radiograph showing bilateral metastatic medullary nephrocalcinosis in a patient with distal renal tubular acidosis.

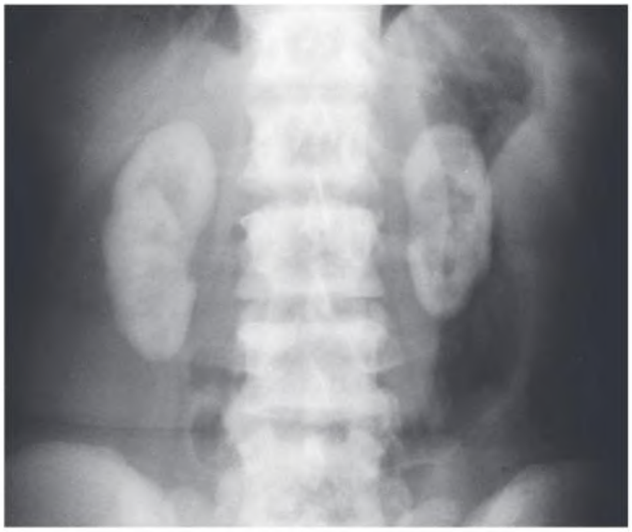

**Fig. 60.9** **Nephrocalcinosis.** Dense cortical and medullary calcification in the shrunken kidneys of a patient with oxalosis and long-standing kidney failure.

medullary sponge kidney, as well as medications, including acetazolamide, amphotericin, and triamterene (see Box 60.3).

In children, a similar range of disorders can be seen, but the most common associations are with furosemide and the hereditary disorders associated with hypercalciuria.[24] Furosemide, when used in premature neonates and older infants with congestive heart failure, can result in nephrocalcinosis with or without apparent hypercalciuria. The lesions often resolve with discontinuation of therapy. A normal ratio of calcium to creatinine at the time of diagnosis of nephrocalcinosis (~0.40 mg/mg in premature infants) appears to be a good predictor of resolution.

There are many uncommon hereditary disorders associated with nephrocalcinosis, including Dent disease, X-linked hypophosphatemic rickets, hypomagnesemia-hypercalciuria syndrome, and Bartter syndrome.

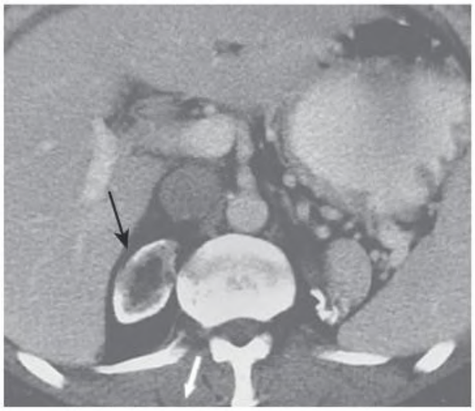

**Fig. 60.10** **Cortical Nephrocalcinosis.** Noncontrast computed tomgraphy scan showing cortical nephrocalcinosis *(arrows)* in the right kidney after cortical necrosis.

Dent disease is an X-linked recessive disorder of proximal tubules characterized by low-molecular-weight proteinuria with hypercalciuria and nephrocalcinosis.[60] In most cases, it is due to mutations that inactivate a voltage-gated chloride transporter called *CLC-5*. A number of mutations affecting the *CLCN5* gene on the X chromosome have been identified that lead to inactivation of CLC-5. The result is a clinical syndrome typically affecting young boys and usually including hypercalciuria, nephrocalcinosis, nephrolithiasis, and hematuria, as well as low-molecular-weight proteinuria, glycosuria, aminoaciduria, hypophosphatemia, kidney failure, and rickets. Dent disease is further discussed in Chapter 50.

In X-linked hypophosphatemic rickets, the traditional treatment, with phosphate repletion and vitamin D, may itself result in hypercalcemia, hypercalciuria, and nephrocalcinosis. Recently, the monoclonal immunoglobulin antibody burosumab, which binds excess fibroblast growth factor 23, was approved for the treatment of this disorder. The drug normalized phosphorus levels in adults and children, and its use should prevent the calcific disorders caused by X-linked hypophosphatemic rickets.

Another cause of medullary nephrocalcinosis in children is primary hypomagnesemia-hypercalciuria syndrome.[3,61] This rare autosomal recessive condition results from defective production of the cellular tight-junction protein paracellin-1. This claudin family protein is necessary for adequate calcium and magnesium reabsorption in the thick ascending limb of the loop of Henle. Children typically present with symptoms of UTI (often with nephrolithiasis), polyuria, tetanic seizures (caused by hypomagnesemia), and muscle cramps and weakness. Hypercalciuria, hypermagnesuria, and a urinary concentrating defect also occur. Patients often have reduced GFR and may require kidney replacement therapy by the third decade of life. Sensorineural hearing disorders and ocular impairment may accompany the kidney manifestations in a subset of patients.

## Cortical Nephrocalcinosis

Cortical nephrocalcinosis is usually the result of dystrophic calcification, which follows parenchymal tissue destruction, rather than the precipitation of excessive urinary constituents. It is secondary to infarction, neoplasm, and infection. It is typically asymmetric and is usually localized to the renal cortex (Fig. 60.10). Causes of cortical nephrocalcinosis include transplant rejection, cortical necrosis (mostly following hemorrhagic shock), tuberculosis, ethylene glycol toxicity, and chronic glomerulonephritis.

## Clinical Manifestations

Patients who do not have nephrolithiasis associated with nephrocalcinosis are often asymptomatic. Ultrasound and CT scanning are sensitive diagnostic tests for both cortical and medullary nephrocalcinosis, demonstrating the parenchymal calcifications before they can be visualized on plain radiographs. The extent of calcification correlates poorly with GFR.

## Treatment

Similar to nephrolithiasis, treatment of nephrocalcinosis relies on therapy for the underlying disease, as well as measures to reduce hypercalcemia, hyperphosphatemia, and oxalosis, if possible. The goal of treatment is usually to prevent further deposits, because therapy cannot eradicate existing calcium deposits.

# SELF-ASSESSMENT QUESTIONS

1. A 49-year-old paraplegic man is referred for evaluation of nephrolithiasis. He has a suprapubic catheter and has had frequent urinary tract infections. He has reduced kidney function and is noted to have large staghorn calculi in both kidneys. On urinalysis, he has many white blood cells and bacteria. Urine culture grows *Proteus.* All statements regarding this patient's kidney stones are true *except:*
   - **A.** His current stones can be treated with a course of antibiotics.
   - **B.** Stone formation is facilitated by the effects of urease-producing bacteria.
   - **C.** His stones are formed by a combination of phosphate with three cations.
   - **D.** Stone formation is potentiated by an elevated urine pH.
   - **E.** His current stones require surgical management.
2. A 35-year-old woman presents with her fourth kidney stone. Her father and brother have had stones as well. She was informed that her stones are composed of calcium. She tries to drink as much fluid as possible. Urine calcium excretion is elevated, but serum calcium is normal. All of the following dietary measures may be recommended for kidney stone prevention in this patient *except:*
   - **A.** Low-sodium diet
   - **B.** Reduction in animal protein intake
   - **C.** Reduction in dietary calcium intake
   - **D.** Low-oxalate diet
3. A 52-year-old obese man presents with flank pain and gross hematuria. Computed tomography reveals an obstructing left kidney stone and two other nonobstructing kidney stones in the right kidney measuring 0.5 mm. He passes the left kidney stone, which is found on analysis to be a uric acid stone. He was recently started on metformin for diabetes mellitus, his body mass index is 33, and he has normal kidney function; 24-hour urine reveals urine pH 5.3, volume 1.5 L, and uric acid excretion of 500 mg. Which of the following is a *true* statement regarding this patient's kidney stones?
   - **A.** The patient's right kidney stones can be followed by serial abdominal radiographic examination of the kidneys.
   - **B.** The patient's insulin resistance may be contributing to his stone disease.
   - **C.** The patient should be treated with allopurinol.
   - **D.** Potassium citrate would not be beneficial.
   - **E.** The patient is drinking an adequate amount of fluid.

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